Dr Muhammad Waqas

Graduate Work Visa | Muhammad Waqas | Muhammad Waqas
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Summary

Materials Scientist with over 9 years of experience in polymer science, sustainable materials, composites, biomaterials, and manufacturing process development. Led multidisciplinary research and development projects from concept through validation, supporting product development, process optimisation, scale-up, and commercial feasibility. Experienced in experimental design, materials characterisation, technical documentation, and stakeholder engagement across academic and industrial environments. Delivered projects with industry partners, developed testing protocols, and translated research outcomes into practical manufacturing solutions. Certified ISO 9001:2015 Lead Auditor with expertise in quality systems, risk management, root cause analysis, and continuous improvement.

Work Experience

Research Fellow (Sprout Smart – Innovate UK) April 2025 - Present

Edinburgh Napier University, Edinburgh, UK

- Developed biodegradable polymer formulations and electrospun carriers for controlled nutrient release systems using bio-based polymers and green solvents.
- Manage project delivery across academic and industrial partners, supporting KPIs, risk management, and commercial objectives.
- Evaluated rheological, thermal, and mechanical behaviour of polymer systems using FTIR, DSC, TGA, and rheology techniques.
- Coordinated with industry (Impact Solutions) on specification, performance targets, environmental profile, and route-to-market assumptions.
- Generated testing protocols and data packages to inform TRL advancement and downstream validation.
- Prepared documentation aligned to REACH considerations and biodegradability screening, supported IP and licensing discussions.

Visiting Researcher April 2024 - November 2024

The University of Edinburgh, Edinburgh, UK

- Designed and developed carbon fibre-reinforced polymer composites embedded with sensing systems, contributing to enhanced material characterisation and Digital Twin (DT) model advancements.
- Recycled low-value packaging waste into composites via pre-compounding mixed plastics with LDPE/CNT using melt-extrusion.

PhD Researcher February 2018 - July 2023

The University of Edinburgh, Edinburgh, UK

- Lead research in developing advanced polymer materials for aerospace and automotive applications.
- Led full characterisation workflows of polymers, nanomaterials, and composites using XRD, TGA, DSC, FTIR, FIB-SEM, SEM, EDX, DMTA, DIC, and mechanical testing.
- Developed high-throughput manufacturing equipment and validate proof-of-concept technologies.
- Generated reproducible datasets and maintain detailed laboratory records and technical documentation.
- Conducted COSHH and risk assessments to ensure a safe and regulatory-compliant environment for handling chemicals, metals, and coatings.
- Presented research papers at prestigious international conferences and workshops.

Quality Control Engineer April 2016 - January 2018

FAUZ Engineering Ltd., Lahore, Pakistan

- Led a team of four to complete 16 industrial projects, achieving compliance with ASME, ASTM, and ISO 9001 standards, which enhanced client satisfaction by 20%.
- Liaised with customers and cross-functional teams to meet project requirements.
- Produced materials testing, failure investigations, technical reports, inspection procedures, testing protocols, and quality documentation.
- Conducted materials testing, failure investigations, welding inspections, and validation activities.

Education

2018 – 2023 **PhD Materials Science and Engineering**
The University of Edinburgh **Principal Supervisor: Prof. Vasileios Koutsos**
Thesis Title: Electrospun Nanofibre Veils and Carbon Fibre-based Veils for Enhanced Electrical Conductivity of Carbon Fibre Reinforced Polymer Composites.

2014 – 2017 **MSc Metallurgical and Materials Engineering**
University of Engineering and Technology, Lahore, Pakistan
Thesis Title: Synthesis and Characterization of Antibacterial Zn doped Hydroxyapatite by Microwave Irradiation

2008 – 2013 **BSc Metallurgy and Materials Engineering**
University of the Punjab, Lahore, Pakistan
Project Title: Failure Analysis of Mild Steel Cylinder Filled with Anhydrous Ammonia

Skills

Analytical Techniques	XRD, FTIR, FIB-SEM, SEM, EDX, DMTA, DSC, Metallography, Mechanical Testing, Optical Profilometry, Electrical Impedance Spectroscopy
Technical Skills	Electrospinning, Encapsulation & Release Kinetics, Root Cause Analysis, Process Failure Mode Effects Analysis, ISO 9001 QMS, Hardness/microhardness Testing, Additive Manufacturing, 3D Printing
Laboratory Management:	Equipment Maintenance, COSHH Compliance, Risk Assessments, Health & Safety Protocols
Software Skills	MS Office, Computer-Aided Design, ImageJ, Finite Element Analysis (FEA), Origin Pro, One Drive, Online Teaching Tools (Teams, Zoom)
Soft Skills	Scientific Writing & Presentations, Project Management, Stakeholder Management (industry–university), Public Speaking, Student Support and Mentoring

Research Funding, Awards and Achievements

- Awarded internal funding to lead *Enhancing Assessment and Learning for International Students in Diverse Faculty Contexts*, supporting inclusive learning and assessment at Edinburgh Napier University (November 2025–June 2026).
- Secured a scholarship of £107k for pursuing PhD at the University of Edinburgh.
- Master Card Foundation reflection coach on *Identities in transition* for African students (2019-20).
- Lothian Equal Access Programme for Schools (LEAPS) volunteers support & promoting STEM Higher Education in Secondary School students in Southeast Scotland (2022-2023).
- Edinburgh Teaching Awards Mentor, coaching & reflection on the AFHEA essays of PhD students and technicians (June 2021 - September 2023).

- Winner of *SAMPE UK & Ireland Design and Make competition 2022* held at AMRC Sheffield.

Training and courses

- Introduction to Life Cycle Thinking, UN Environment Programme (2025)
- Understanding Business Improvement Techniques (SCQF Level 5), West College of Scotland (2025)
- Carbon Literate in carbon footprints by Carbon Literacy project, UK (2022).
- Ceramics-structural and Advanced application course, Pakistan Engineering Council (2016).

Professional Memberships

- Certified ISO 9001:2015 Lead Auditor, CQI and IRCA
- Associate Fellow in Higher Education Academy (AFHEA) (PR217616)
- Registered Engineer, Pakistan Engineering Council (METAL-3110)

International Journal Publications Muhammad Waqas

Mazhar Hussain, Muhammad Farooq, Muhammad Asad Saeed, Muhammad Zaman, Sherjeel Adnan, Zeeshan Masood, Nabeela Ameer, **Muhammad Waqas**, Hafiz Muhammad Abdur Rahman, and Muhammad Ijaz (2026). "Enhanced Mucoadhesion and Solubility of Aprepitant-Loaded Thiolated Chitosan Solid Lipid Nanoparticles for Sustained Drug Release". In: *ChemistrySelect*. Accepted, in press.

Mazhar Hussain, Muhammad Farooq, Muhammad Asad Saeed, Muhammad Ijaz, Sherjeel Adnan, Zeeshan Masood, **Muhammad Waqas**, Wafa Ishaq, and Nabeela Ameer (2025). "Aprepitant-loaded solid lipid nanoparticles: a novel approach to enhance oral bioavailability". In: *Beilstein Journal of Nanotechnology* 16.1, pp. 652–663.

Kit O'Rourke, Christopher Griffin, Keith Doyle, **Muhammad Waqas**, Paula Douglas, Bronagh Millar, and Dipa Ray (2025). "Trash into Treasure: Value-Added Composites from Waste Plastic Packaging and Carbon Nanotubes". In: *Materials Today Sustainability*, p. 101231.

Muhammad Imran-Masood, Enrique Garcia-Diez, Muhammad Usman, Balaj Khan Lodhi, **Muhammad Waqas**, and Susana Garcia (2024). "Development of a novel bio char for CO2 capture and biogas upgrade: Static and dynamic testing". In: *Journal of CO2 Utilization* 89, p. 102958.

Wafa Ishaq, Attia Afzal, Muhammad Farooq, Muhammad Sarfraz, Sherjeel Adnan, Hammad Ahmed, **Muhammad Waqas**, and Zainab Safdar (2024). "Design and Evaluation of Inorganic/Organic Hybrid Bio-composite for Site-Specific Oral Delivery of Darifenacin". In: *AAPS PharmSciTech* 25.

Muhammad Waqas, Francisco Javier Diaz Sanchez, Valentin C Menzel, Ignacio Tudela, Norbert Radacsi, Dipa Ray, and Vasileios Koutsos (2023). "Polyaniline/polyvinylpyrrolidone nanofibers via nozzle-free electrospinning". In: *Journal of Applied Polymer Science*, e54586.

Muhammad Waqas, Colin Robert, Urwah Arif, Norbert Radacsi, Dipa Ray, and Vasileios Koutsos (2022). "Improving the through-thickness electrical conductivity of carbon fiber reinforced polymer composites using interleaving conducting veils". In: *Journal of Applied Polymer Science* 139.43, e53060.

Francisco Javier Diaz Sanchez, Michael Chung, **Muhammad Waqas**, Vasileios Koutsos, Stewart Smith, and Norbert Radacsi (2022). "Sponge-like piezoelectric micro-and nanofiber structures for mechanical energy harvesting". In: *Nano Energy* 98, p. 107286.

Faraz Fazal, Francisco Javier Diaz Sanchez, **Muhammad Waqas**, Vasileios Koutsos, Anthony Callanan, and Norbert Radacsi (2021). “A modified 3D printer as a hybrid bioprinting-electrospinning system for use in vascular tissue engineering applications”. In: *Medical Engineering & Physics* 94, pp. 52–60.

Muhammad Waqas, Antonios Keirouz, Maria Kana Sanira Putri, Faraz Fazal, Francisco Javier Diaz Sanchez, Dipa Ray, Vasileios Koutsos, and Norbert Radacsi (2021). “Design and development of a nozzle-free electrospinning device for the high-throughput production of biomaterial nanofibers”. In: *Medical Engineering & Physics* 92, pp. 80–87.

Colin Robert, Witiwat Best Thitasiri, Dimitrios Mamalis, Zakareya Elmo Hussein, **Muhammad Waqas**, Dipa Ray, Norbert Radacsi, and Vasileios Koutsos (2021). “Improving through-thickness conductivity of carbon fiber reinforced polymer using carbon nanotube/polyethylenimine at the interlaminar region”. In: *Journal of Applied Polymer Science* 138.5, p. 49749.

International Conference Proceedings Publications

Muhammad Waqas, Dong Sun, and Reza Salehiyan (Apr. 2026). “Electrospun Nanofibrous Scaffolds for Biomedical and Soft Packaging Applications”. In: *Polymers / Composites / 3Bs Materials Tech 2026 Joint International Conferences Preliminary Program*. Lisbon, Portugal.

Muhammad Waqas, Quimm. James A., Sergejs Afanasjevs, Konstantin Kamenev, Norbert Radacsi, Dipa Ray, and Vasileios Koutsos (2024). “Electrospun Polymer Nanofibres for Enhanced Through-Thickness Electrical Conductivity of CFRP Composites”. In: *University of Edinburgh, Physical Aspects of Polymer Science 2024*. Institute of Physics.

Muhammad Waqas and Simone Dimartino (2023). “Teaching assistant account: First year students perception and experience of online, face-to-face teaching during and after COVID-19”. In: *Proceedings of the Learning and Teaching Conference, The University of Edinburgh*.

Wafa Umair, Attia Afzal, Muhammad Farooq, Muhammad Sarfraz, Sherjeel Adnan, Hammad Ahmad, and **Muhammad Waqas** (2022). “Development and characterization of calcium carbonate-quince bio-composite for pH triggered release of darfenacin hydrobromide in lower GIT: A green chemistry approach”. In: *Proceedings of the 2nd International Electronic Conference on Biomolecules*.

Muhammad Waqas, Tahir Ahmad, and Faraz Hussain (2014). “Failure analysis of mild steel cylinder filled with anhydrous ammonia”. In: *Proceedings of the First International Young Engineers Convention, University of Engineering and Technology, Lahore, Pakistan*.